

PRST Class (11/11/2020)

Sampling distribution :

gene profile of all Indians
population $N(\mu, \sigma^2)$
 μ, σ : unknown

sample of size n
(1, ..., n)

X_1 : gene profile of the 1st person in the random sample

$X_i \sim N(\mu, \sigma^2)$

X_2 : ~~gene~~ gene profile of the 2nd person in the random sample.

$\vdots$
 X_n : μ 2nd person

$(X_1, X_2, \dots, X_n)$ $n=100$

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ — sample mean

$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$ — sample variance $\rightarrow$

$\frac{1}{n} \sum_{i=1}^n X_i^3$

Statistic: — any function of the random sample $(X_1, \dots, X_n)$.

$T(X_1, \dots, X_n) = T$

T — has a prob distribution $\rightarrow$ sampling distribution

$SD(T) = [\text{Var}(T)]^{\frac{1}{2}} \rightarrow$ standard error

S_1 S_2 $\dots$ S_{10}
 $X_1, X_2, \dots, X_n$
 T t_1 t_2 t_{10}
 $t = \bar{x}$

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$

$X = \frac{1}{2} \quad V(X)$
 $\vdots$
 4

$T = \frac{1}{n} \sum_{i=1}^n X_i$

$SD(T) = SD\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{\sigma}{\sqrt{n}}$

χ^2 - distribution

$X \sim N(\mu, \sigma^2)$

$\Rightarrow Y = \frac{X - \mu}{\sigma} \sim N(0, 1)$

$Y^2 \sim \chi_0^2$

degrees of freedom = 1

$X_1, \dots, X_n$ independent $\left| \begin{array}{l} X_1 \sim N(\mu, \sigma^2) \\ X_2 \sim N(\mu, \sigma^2) \\ \vdots \\ X_n \sim N(\mu, \sigma^2) \end{array} \right. \quad X = \mu$

$X_1, \dots, X_n$ are independent $X_n \sim N(\mu, \sigma^2)$

$$Y_1 = \frac{X_1 - \mu}{\sigma}, Y_2 = \frac{X_2 - \mu}{\sigma}, \dots, Y_n = \frac{X_n - \mu}{\sigma}$$

$$Y_1^2, Y_2^2, \dots, Y_n^2$$

$$Y_1^2 + Y_2^2 + \dots + Y_n^2 \sim \chi_n^2$$

degrees of freedom = n

$$e^{-\frac{x}{2}} \frac{x^{\frac{n}{2}-1}}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}, n > 0$$

pdf of χ_n^2

$$f(x) = \frac{1}{2^{\frac{n}{2}} \Gamma(\frac{n}{2})}$$

$$\left(\alpha = \frac{1}{2}, \beta = \frac{n}{2} \right)$$

$$\chi_n^2 \equiv \text{Gamma}\left(\frac{1}{2}, \frac{n}{2}\right)$$

$$E(\chi_n^2) = n$$

$$\text{Var}(\chi_n^2) = 2n$$

$$W_1 \sim \chi_{n_1}^2$$

$$W_2 \sim \chi_{n_2}^2$$

where W_1, W_2 are independent

$$W_1 + W_2 \sim \chi_{n_1+n_2}^2$$

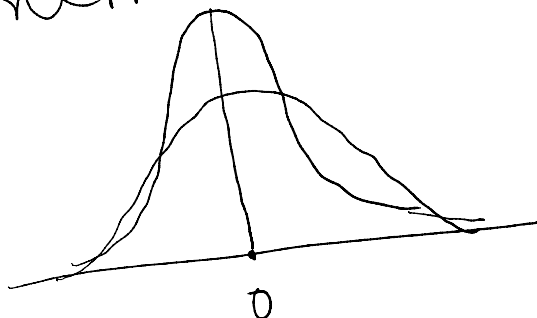
t - distribution

$$X \sim N(0, 1), Y^2 \sim \chi_n^2$$

X and Y^2 are independent

$$t = \frac{X}{\sqrt{\frac{Y^2}{n}}} \sim t_n; \text{ degrees of freedom} = n$$

t dist is symmetric about 0.



$$\frac{1}{\Gamma(n)} \dots \frac{1}{\Gamma(n+1)}$$

$$\text{pdf of } t, f(t) = \frac{\frac{\sqrt{n}}{2} + 1}{\sqrt{n\pi} \sqrt{\frac{n}{2}}} \cdot \frac{1}{\left(1 + \frac{t^2}{n}\right)^{\frac{n+1}{2}}}$$

$$-\infty < t < \infty$$

r th moment ~~$E(t^r)$~~ exists iff $n > r$.